

Handheld Instruments: Low-Cost, Rugged Solution for Field and Lab



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Handheld instruments were traditionally meant for field applications, but these are now being used in the research and development sector as a complementary product to existing benchtop instruments. A handheld device weighs less, is rugged, relatively easy to use, and can reduce your initial expense, maintenance cost and storage space. Majority of the benchtop functions are now available in the portable form, without compromising much on accuracy and precision, which is the icing on the cake.

Before going into how to select the best handheld for a particular application, one has to understand if a handheld device is required at all.

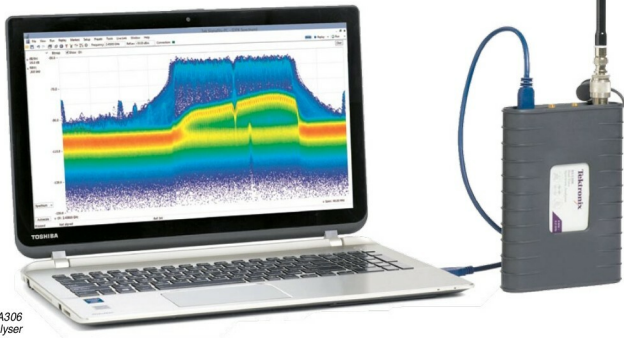
Do I need a handheld

We differentiate handhelds from benchtops primarily based on size, portability, accuracy, features, ease-of-use, cost and ruggedness. If a light-weight, low-cost, small form factor, rugged instrument with

a simple user interface (UI) is what you are looking for, and if the measurement to be taken need not be the most accurate, then you can go for a handheld device. These provide better cost performance.

With traditional instruments the only way to reduce cost was to compromise on quality, that is, use cheaper components. However, with software-defined instruments, it is possible to reduce the cost and size without compromising on quality of measurements. This is one of the reasons why handheld instruments these days are not limited to field usage; these are being used in research and development labs too.

In the field. Portable instruments are mainly used by engineers or technicians in field scenarios for installation, maintenance and monitoring application of systems. This does not generally require the most advanced functionalities or the most precise readings. What is needed is a handy, light-weight, battery-operated device that has the basic capabilities (some-



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